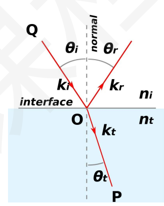
底层视觉领域的一些最新任务

- 镜头相关
  - 几何光学 → 波动光学、误差建模、畸变校正
- 传感器相关
  - 噪声建模、多传感器融合
- 新任务
  - 去恶劣天气、去反光
- 新目标
  - 人脸图像、文本图像

镜头

光学

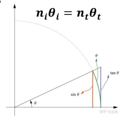
- Geometry Optics (几何光学) & Wave Optics (波动光学)

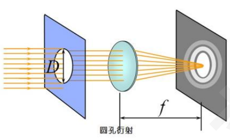


- 斯涅耳定律 (Snell's Law)
  $n_i \sin\theta_i = n_t \sin\theta_t$
- 小角度近似 (5° 误差约1%)
  $\sin\theta \approx \tan\theta \approx \theta$

i.e.,

$n_i\theta_i = n_t\theta_t$





圆孔衍射 (透镜衍射类似)
属于菲涅尔衍射 (Fresnel Diffraction)

https://zhuanlan.zhihu.com/p/354346141
https://baike.baidu.com/item/%E5%85%89%E7%9A%84%E8%A1%8D%E5%B0%84/1019091?fr=aladdin

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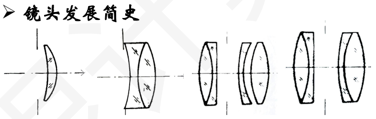
镜头（几何光学） 不断校正各种"像差"，让成像更清晰、更真实

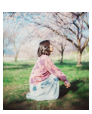
- 镜头发展简史

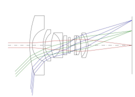


1812 新月

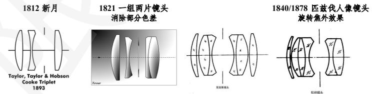
1821 一组两片镜头
消除部分色差

1840/1878 匹兹伐人像镜头
旋转焦外效果





Code V 鱼眼镜头示例



1893 库克
大光圈、大相场、小体积，现多用于电影镜头

1902 天塞
凹透镜矫正库克像差
成像锐利，适合黑白摄影

1896 双高斯
大光圈标准镜头标配
较好地矫正各类像差

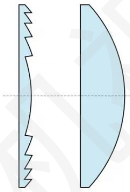
1929 松纳
（双高斯+天塞）
标准到中长焦镜头
球差小，光圈极大，色彩鲜亮，四角失光少

佳能超长焦

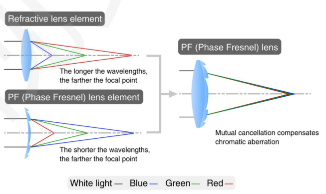
https://zhuanlan.zhihu.com/p/356933838
https://baijiahao.baidu.com/s?id=1727981058211381463

其他镜头

- Fresnel Lens (菲涅尔透镜， Diffractive optical elements, DOE)



菲涅尔透镜 vs 普通透镜



利用光衍射现象补偿色差原理

优势：利用光衍射现象补偿色差，大幅减少体积与焦距
缺点：(存在高阶衍射导致）画质下降、点光源会形成有色眩光

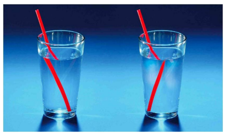
尼康 AF-S NIKKOR 300mm f/4E PF ED VR

https://www.bilibili.com/read/cv15206443/
https://www.zhihu.com/question/27432342/answer/36629328

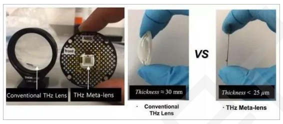
其他镜头

➢ MetaLens（超透镜）

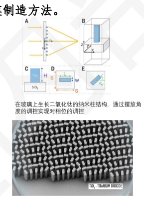
2016: 超透镜提出
2018: 无色差超透镜 (配对纳米膜控制不同波长光的速度)
2020: 结构表面+液晶 (通过液晶实现可重构性)
2022: iPhone 高端机型激光雷达商用
Future (Dielectric Metalens for Miniaturized Imaging Systems: Progress and Challenges. *Light: Sci & App, 2022.*):
1 打破超透镜性能之间冲突的新策略；
2 多功能和可重新配置的超透镜，可以取代传统光学的复杂配置。
3 高性能超透镜构件的高效率设计方法 。
4 大面积($d = 0.5mm$)超透镜的高效率制造和大规模制造方法。



负折射率超材料
2011 Science，广义折反射定律



超透镜（光刻）
2016 Science 封面论文



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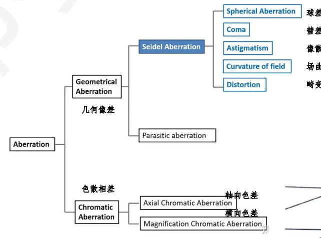
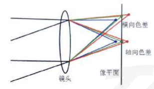
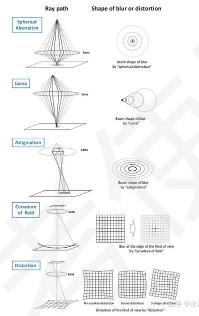
https://zhuanlan.zhihu.com/p/339440254
https://zhuanlan.zhihu.com/p/578596624
https://mp.weixin.qq.com/s?__biz=Mzg3MTY2NjY1OA==&mid=2247516691&idx=4&sn=946b6e465cb365c5a81f9b37b808b9ee&chksm=cef9c49ef98e4d88058f431caa5548d93dd83f28a2c6f9f637d46ac67ecacddbb9540e5180f4&scene=27

Aberration（像差）

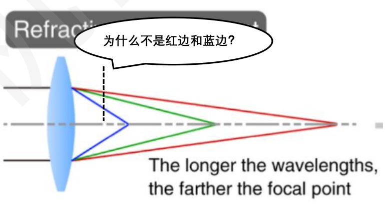




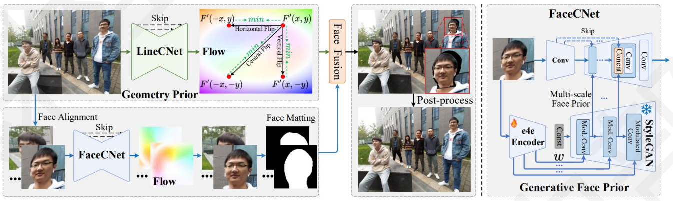


https://zhuanlan.zhihu.com/p/357815034

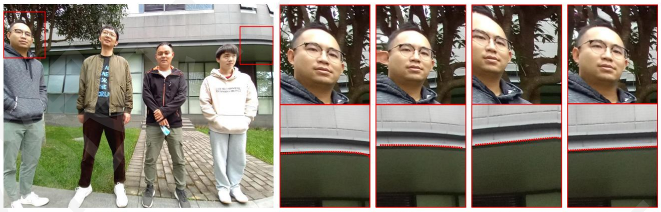
像散色差（绿边、紫边）



广角畸变校正



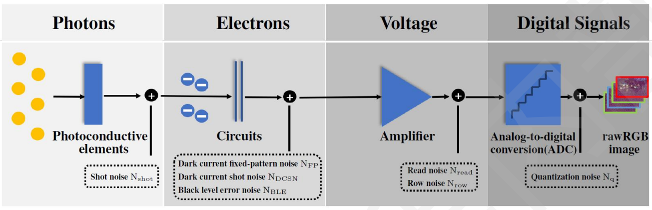
广角畸变校正



Full input image | Close-up of input | Shih's *et al* | Zhu's *et al* | Ours

传感器

噪声建模



噪声建模

- 散粒噪声

$$N_{shot} \sim \mathcal{N}(0, \beta_{shot} \cdot I)$$

- 暗电流噪声

$$N_{DC} = k \cdot N_{FP} + N_{BLE} + N_{DCSN}$$

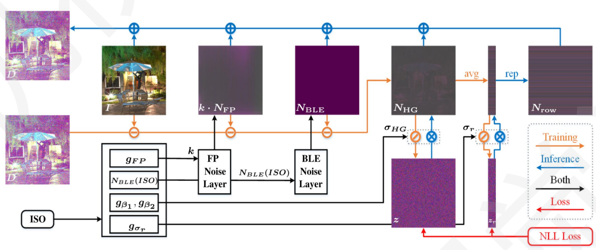
- 读出噪声

$$N_{read} \sim \mathcal{N}(0, \sigma_{read}^2), N_{row} \sim \mathcal{N}(0, \sigma_{row}^2)$$

- 量化噪声

$$N_q \sim U(-\frac{1}{2}q, \frac{1}{2}q)$$

噪声建模



噪声建模方法对比

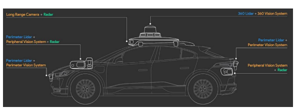
方法	类别	散粒噪声 Shot Noise	固定模式噪声 Fix-pattern Noise	黑电平误差噪声 Black Level Error Noise	暗电流散粒噪声 Dark Current Shot Noise	读出噪声 Read Noise	行噪声 Row Noise	量化噪声 Quantization Noise	可学习性	ISO相关性
ELLE	标定	✓		✓	✓	✓	✓	✓	☆☆☆☆☆	★★★☆☆
ELD	标定	✓		采样	✓	✓	✓	✓	☆☆☆☆☆	★★★☆☆
SFRN	标定	✓	采样	采样	采样	采样	采样	✓	☆☆☆☆☆	★★★☆☆
PMN	标定	✓	✓	✓				✓	☆☆☆☆☆	★★★★☆
NoiseFlow	数据驱动	✓			✓	✓		✓	★★★★★	★★☆☆☆
Starlight	数据驱动	✓	✓		✓	✓	✓	✓	★★★☆☆	☆☆☆☆☆
LLD	数据驱动	✓	✓	✓	✓	✓	✓	✓	★★★★★	★★★★★

多摄/多曝增强

多摄多传感器

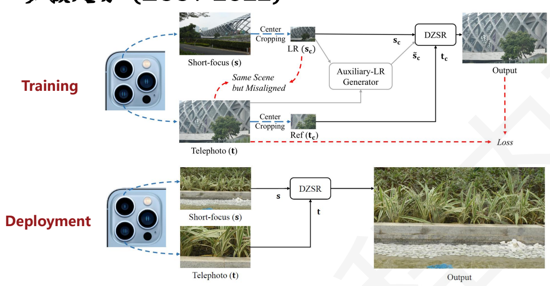
Apple Vision Pro (14 Cameras)

iPhone 15

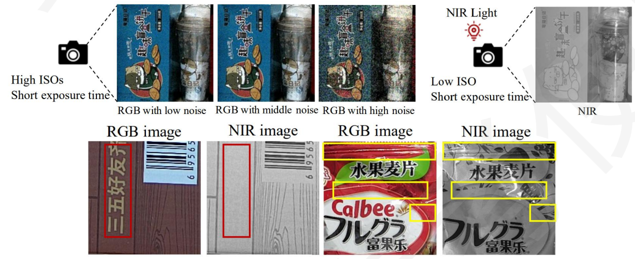


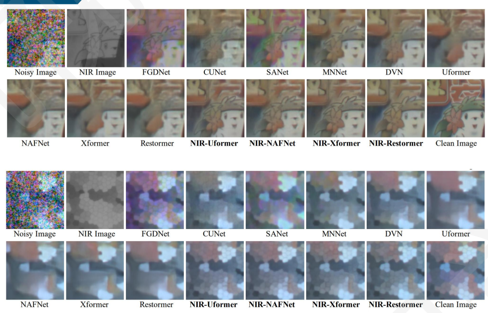
Waymo

多摄超分（ECCV 2022）

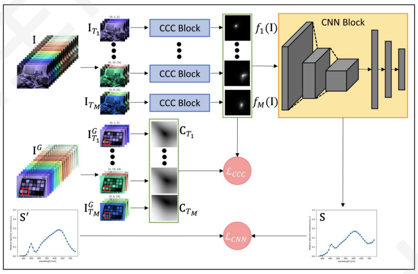


可见光-近红外融合

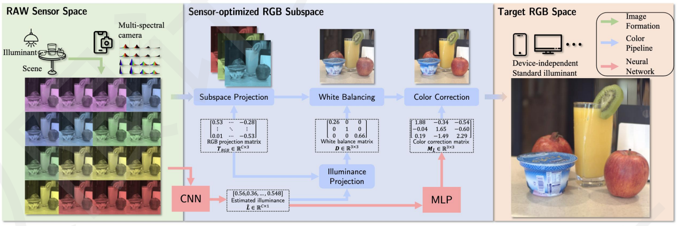




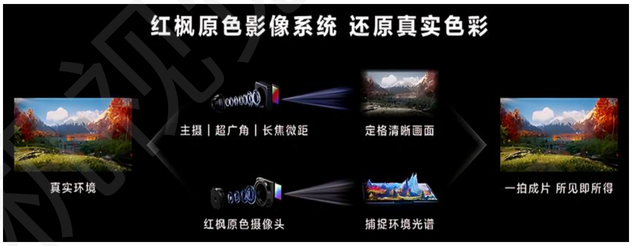
可见光-多光谱融合 – 多光谱光源估计与白平衡



可见光-多光谱融合 – 多光谱成像



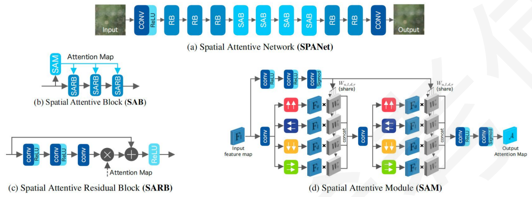
可见光-多光谱融合 – 多光谱成像



去恶劣天气/去反光

SPA (Semantic-guided Pixel-level Attention ， CVPR 2019)

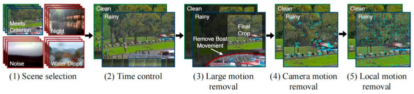
- 构建了一个大规模的成对去雨数据集，主要提出了一种从真实序列帧中估计出一张高质量清晰图的方法
- 提出了SPANet to remove rain streaks in a local-to-global manner



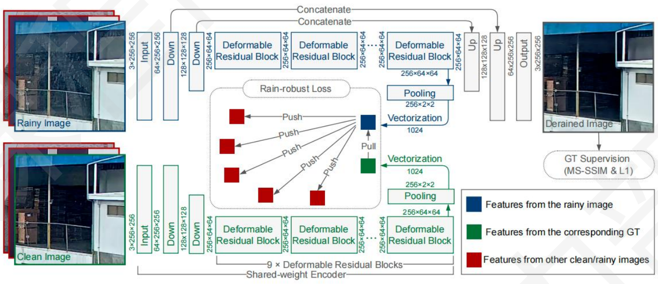
Not Just Streaks（ECCV2022）

GT-RAIN数据集

Dataset	Type	Rain Effects	Size
Rain12 [29]	Simulated	Synth. streaks only	12
Rain100L [50]	Simulated	Synth. streaks only	300
Rain800 [57]	Simulated	Synth. streaks only	800
Rain100H [50]	Simulated	Synth. streaks only	1.9K
Outdoor-Rain [27]	Simulated	Synth. streaks & Synth. accumulation	10.5K
RainCityscapes [19]	Simulated	Synth. streaks & Synth. accumulation	10.62K
Rain12000 [56]	Simulated	Synth. streaks only	13.2K
Rain14000 [12]	Simulated	Synth. streaks only	14K
NYU-Rain [27]	Simulated	Synth. streaks & Synth. accumulation	16.2K
SPA-Data [44]	Semi-real	Real streaks only	29.5K
Proposed	Real	Real streaks & Real accumulation	31.5K

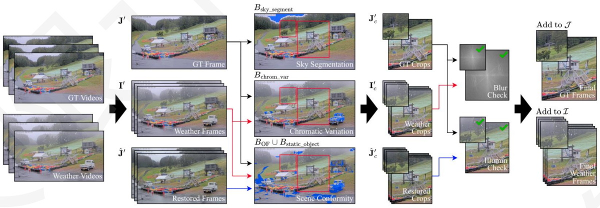


Not Just Streaks（ECCV2022）

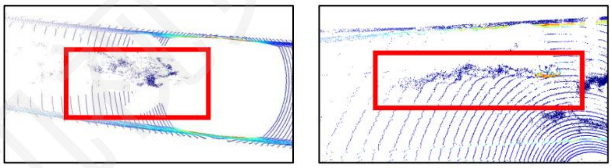


WeatherStream（CVPR 2023）

- 利用光传输技术，将其 从人为筛选变为自动收集真实图像对
- 背景一致（Background Conformity），粒子色差（Particle Chromatic Variation），基于散射的模糊（ Scatterdependent Blur），光照一致（Illumination Consistency）

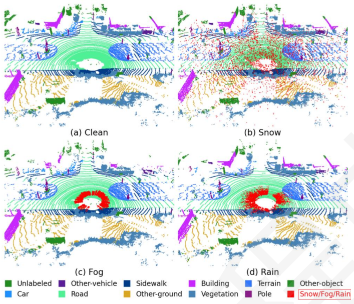


激光雷达去雨/去雪/去雾

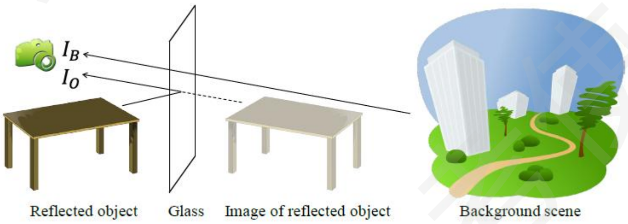


雨天车辆溅射现象

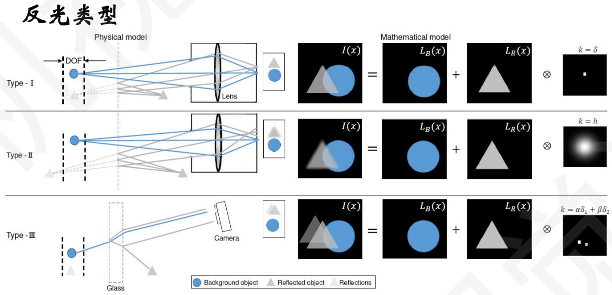
激光雷达去雨/去雪/去雾



去反光



反光类型



典型的反光数据合成方法

Randomly pick two natural images normalized to $[0,1]$ as background $\mathbf{B}$ and reflection $\mathbf{R}$ respectively, then:

1. $\tilde{\mathbf{R}} \leftarrow gauss\_blur_\sigma(\mathbf{R})$ with $\sigma \sim \mathcal{U}(2,5)$
2. $\mathbf{I} \leftarrow \mathbf{B} + \tilde{\mathbf{R}}$
3. $m \leftarrow mean(\{\mathbf{I}(\mathbf{x},c) \mid \mathbf{I}(\mathbf{x},c) > 1, \forall \mathbf{x}, \forall c = 1,2,3\})$
4. $\tilde{\mathbf{R}}(\mathbf{x},c) \leftarrow \tilde{\mathbf{R}}(\mathbf{x},c) - \gamma \cdot (m-1), \forall \mathbf{x}, \forall c$; γ set as 1.3
5. $\tilde{\mathbf{R}} \leftarrow clip_{[0,1]}(\tilde{\mathbf{R}})$
6. $\mathbf{I} \leftarrow clip_{[0,1]}(\mathbf{B} + \tilde{\mathbf{R}})$

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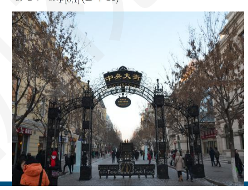
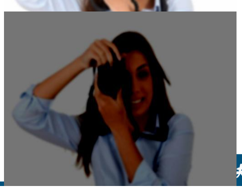
6. $\mathbf{I} \leftarrow clip_{[0,1]}(\mathbf{B} + \tilde{\mathbf{R}})$





典型的反光数据合成方法

Randomly pick two natural images normalized to $[0,1]$ as background $\mathbf{B}$ and reflection $\mathbf{R}$ respectively, then:

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2. $\mathbf{I} \leftarrow \mathbf{B} + \tilde{\mathbf{R}}$
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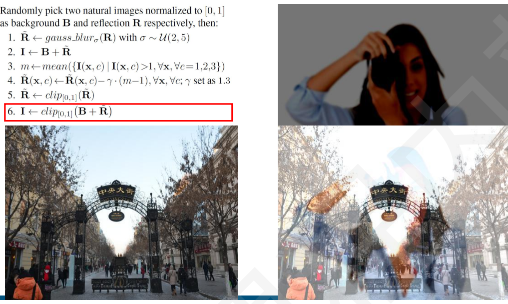
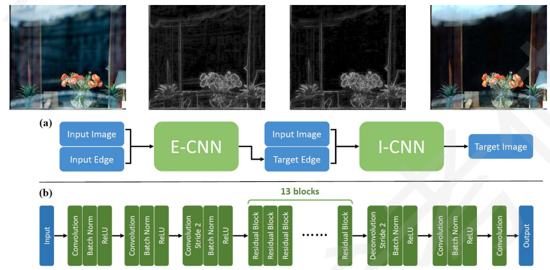
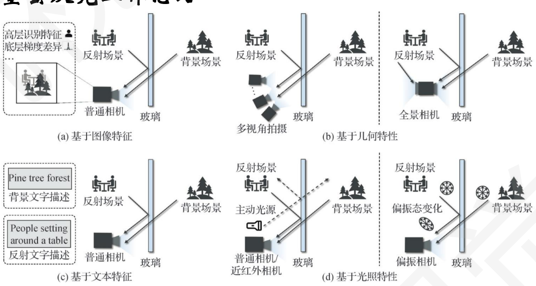
6. $\boxed{\mathbf{I} \leftarrow clip_{[0,1]}(\mathbf{B} + \tilde{\mathbf{R}})}$



典型去反光工作思路



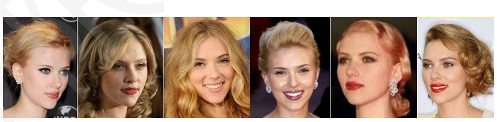
典型去反光工作思路



人脸/文字复原

引导人脸复原

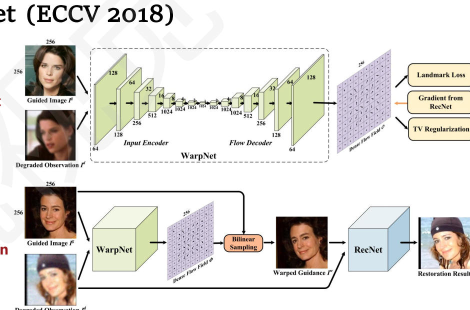
- 盲复原仍有较大难度
- 名人、熟人的人脸盲复原
  - 有一定数量的高质量引导图像
  - 人脸的结构化特点，更适合利用 **生成式先验**



GFRNet（ECCV 2018）

Alignment

Restoration

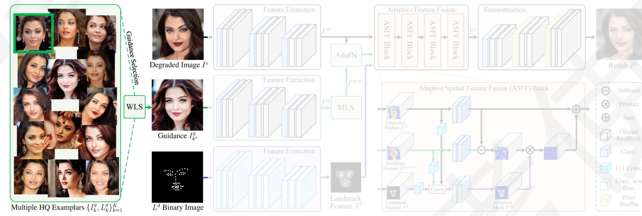


GFRNet (ECCV 2018)

MDnCNN MARCNN MDeblurGAN Ours **Limitation**

哈爾濱工業大學 HARBIN INSTITUTE OF TECHNOLOGY 成像过程中的底层视觉任务 生成模型 – GAN

ASFFNet (CVPR 2020)



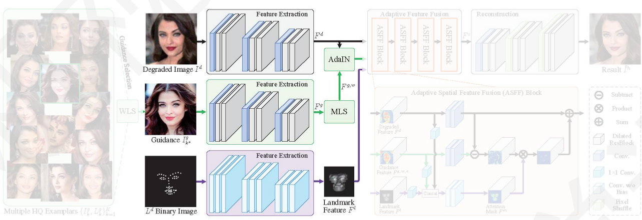
1. Landmark-based Guidance Selection

$$k^*=\arg\min_k \left\{D_a^2(L^d,L_k^g)=\min_A \sum_{m=1}^{68} w_m \left\|A\tilde{L}^g_{k,m}-L^d_m\right\|^2\right\}$$

规格严格 功夫到家

哈爾濱工業大學 HARBIN INSTITUTE OF TECHNOLOGY 成像过程中的底层视觉任务 生成模型 – GAN

ASFFNet (CVPR 2020)



2. MLS for Alignment 和 AdaIN for Illumination Normalization

$$M_p = L^g W_p \tilde{L}^{d\,T}(\tilde{L}^d W_p \tilde{L}^{d\,T})^{-1}$$

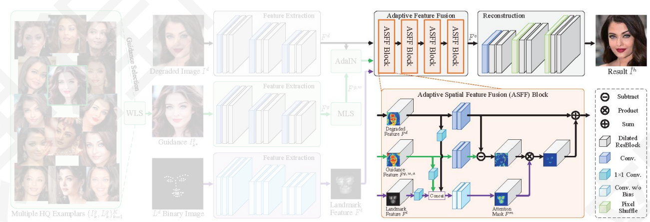
$$F^{g,w}(x,y)=\sum_{(x',y')\in N} F^g(x',y')\max(0,1-|\hat{x}-x'|)\max(0,1-|\hat{y}-y'|)$$

$$F^{g,w,a}=\sigma(F^d)\left(\frac{F^{g,w}-\mu(F^{g,w})}{\sigma(F^{g,w})}\right)+\mu(F^d)$$

规格严格 功夫到家

哈爾濱工業大學 HARBIN INSTITUTE OF TECHNOLOGY 成像过程中的底层视觉任务 生成模型 – GAN

ASFFNet (CVPR 2020)



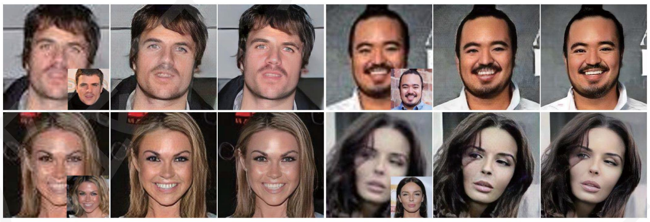
3. Adaptive Feature Fusion

$$\mathcal{F}_{\mathrm{ASFF}}(F^d,F^{g,w,a})=\mathcal{F}_d(F^d)+F^m\circ(\mathcal{F}_g(F^{g,w,a})-\mathcal{F}_d(F^d))$$

规格严格 功夫到家

哈爾濱工業大學 HARBIN INSTITUTE OF TECHNOLOGY 成像过程中的底层视觉任务 生成模型 – GAN

ASFFNet (CVPR 2020)



Input & Guidance GFRNet [24] Ours Input & Guidance GFRNet [24] Ours

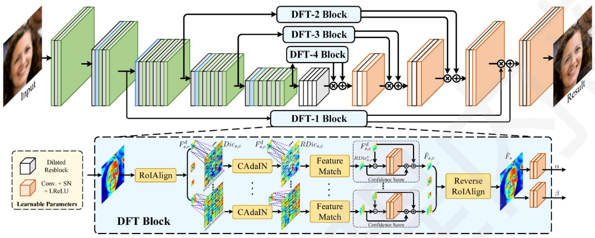
规格严格 功夫到家

哈爾濱工業大學 HARBIN INSTITUTE OF TECHNOLOGY 成像过程中的底层视觉任务 生成模型 – GAN

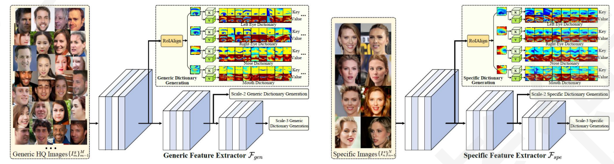
ASFFNet (CVPR 2020)

规格严格 功夫到家

DFDNet（ECCV 2020）

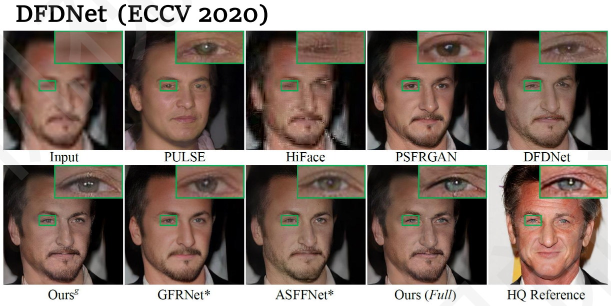


DFDNet（ECCV 2020）



- 同时保留通用字典和个性化字典
- 如果没有个性化字典，则使用通用字典
- 如果有个性化人脸字典，优先使用个性化字典，兼顾使用通用字典

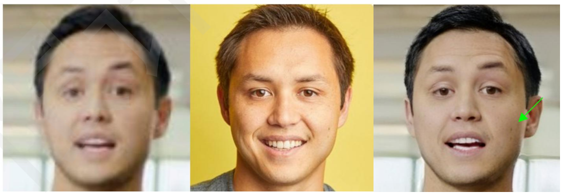
DFDNet（ECCV 2020）



DFDNet（ECCV 2020）

引导人脸复原

Input　Reference　Output

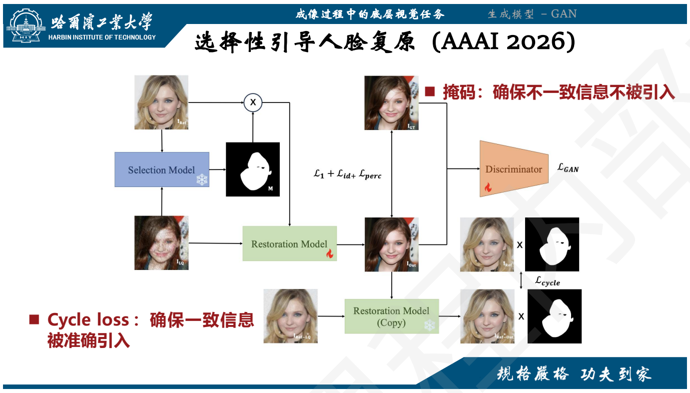


如何确保需需要引入的引导图特征能够准确引入

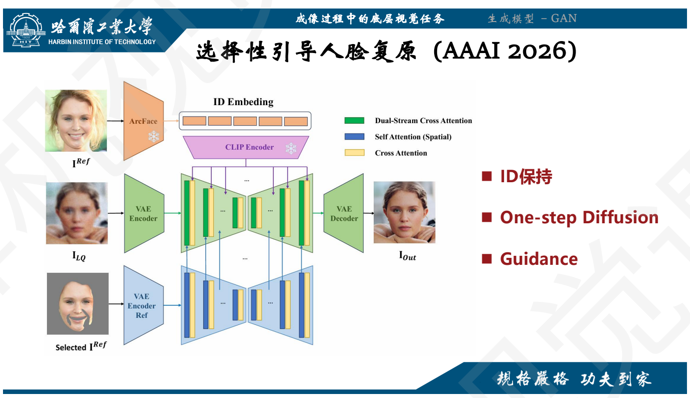
引导人脸复原

Input　Reference　Output

如何避免不需要引入的引导图特征能够不被引入


哈爾濱工業大學
HARBIN INSTITUTE OF TECHNOLOGY
成像过程中的底层视觉任务
生成模型 - GAN
选择性引导人脸复原（AAAI 2026）
Selection Model
Restoration Model
Restoration Model (Copy)
Discriminator
$\mathcal{L}_1 + \mathcal{L}_{id} + \mathcal{L}_{perc}$
$\mathcal{L}_{GAN}$
$\mathcal{L}_{cycle}$
■ 掩码：确保不一致信息不被引入
■ Cycle loss：确保一致信息被准确引入
规格严格 功夫到家


哈爾濱工業大學
HARBIN INSTITUTE OF TECHNOLOGY
成像过程中的底层视觉任务
生成模型 - GAN
选择性引导人脸复原（AAAI 2026）
Input
Ref
Selected Mask
规格严格 功夫到家


哈爾濱工業大學
HARBIN INSTITUTE OF TECHNOLOGY
成像过程中的底层视觉任务
生成模型 - GAN
选择性引导人脸复原（AAAI 2026）
ID Embeding
ArcFace
CLIP Encoder
Dual-Stream Cross Attention
Self Attention (Spatial)
Cross Attention
VAE Encoder
VAE Decoder
VAE Encoder Ref
I^{Ref}
I_{LQ}
I_{Out}
Selected I^{Ref}
■ ID保持
■ One-step Diffusion
■ Guidance
规格严格 功夫到家

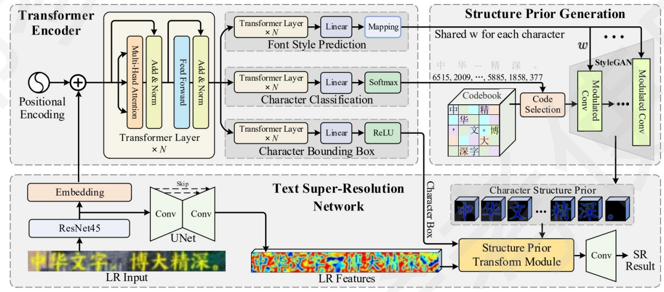

哈爾濱工業大學
HARBIN INSTITUTE OF TECHNOLOGY
成像过程中的底层视觉任务
生成模型 - GAN
选择性引导人脸复原（AAAI 2026）
Input
Ref
Selected Ref
Output
引导细节的准确引入，并有效抑制不一致细节的引入
规格严格 功夫到家


哈爾濱工業大學
HARBIN INSTITUTE OF TECHNOLOGY
成像过程中的底层视觉任务
生成模型 - GAN
选择性引导人脸复原（AAAI 2026）
Input
Selected Ref
Output
引导细节的准确引入
规格严格 功夫到家

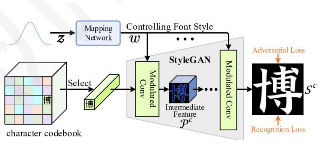

哈爾濱工業大學
HARBIN INSTITUTE OF TECHNOLOGY
成像过程中的底层视觉任务
生成模型 - GAN
选择性引导人脸复原（AAAI 2026）
Input
Output
Selected Ref
有效保证非必要引导图特征（如法令纹）不被引入
规格严格 功夫到家

文本复原

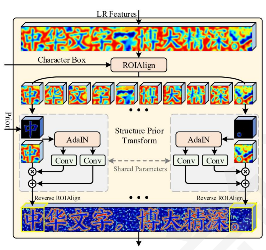
文本复原



文本复原



文本先验学习

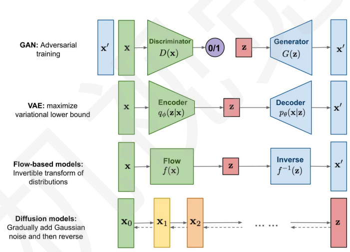


生成/复原特征融合

文本复原

视觉生成模型

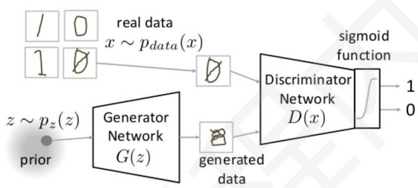
生成模型



GAN	不稳定，且可能导致模式崩溃
VAE	依赖替代目标（变分下界）来近似最大似然训练
Flow	对模型架构有严格限制，以确保似然计算中的归一化常数可处理
DM	效率较低

Generative Adversarial Networks (Goodfellow et al., NIPS 2014)

- **更新生成器**，使其生成更真实的图像
- **更新判别器**，使其能够区分合成图像与真实图像

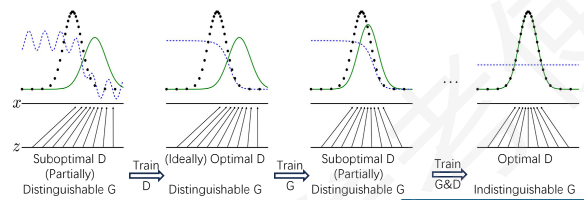


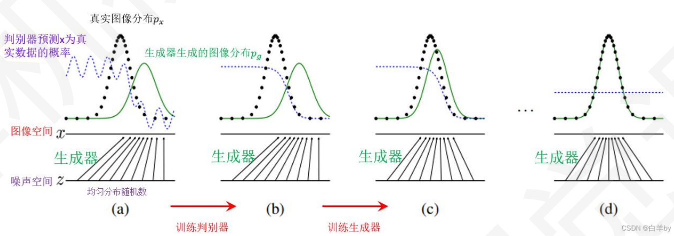
Two-player minimax game

Why log?

- 价值函数（目标函数）

$$\min_G \max_D V(D,G) = \mathbb{E}_{\boldsymbol{x}\sim p_{\text{data}}(\boldsymbol{x})}[\log D(\boldsymbol{x})] + \mathbb{E}_{\boldsymbol{z}\sim p_{\boldsymbol{z}}(\boldsymbol{z})}[\log(1 - D(G(\boldsymbol{z})))].$$





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Algorithm 1 Minibatch stochastic gradient descent training of generative adversarial nets. The number of steps to apply to the discriminator, k, is a hyperparameter. We used $k = 1$, the least expensive option, in our experiments.

for number of training iterations **do**
  for k steps **do**
  - Sample minibatch of m noise samples $\{\boldsymbol{z}^{(1)}, \ldots, \boldsymbol{z}^{(m)}\}$ from noise prior $p_g(\boldsymbol{z})$.
  - Sample minibatch of m examples $\{\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)}\}$ from data generating distribution $p_{\text{data}}(\boldsymbol{x})$.
  - Update the discriminator by ascending its stochastic gradient:

$$\nabla_{\theta_d} \frac{1}{m} \sum_{i=1}^{m} \left[\log D\left(\boldsymbol{x}^{(i)}\right) + \log\left(1 - D\left(G\left(\boldsymbol{z}^{(i)}\right)\right)\right)\right].$$

  end for
  - Sample minibatch of m noise samples $\{\boldsymbol{z}^{(1)}, \ldots, \boldsymbol{z}^{(m)}\}$ from noise prior $p_g(\boldsymbol{z})$.
  - Update the generator by descending its stochastic gradient:

$$\nabla_{\theta_g} \frac{1}{m} \sum_{i=1}^{m} \log\left(1 - D\left(G\left(\boldsymbol{z}^{(i)}\right)\right)\right).$$

end for
The gradient-based updates can use any standard gradient-based learning rule. We used momentum in our experiments.

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end for
The gradient-based updates can use any standard gradient-based learning rule. We used momentum in our experiments.

理论分析：全局最优性

Proposition 1. *For G fixed, the optimal discriminator D is*

$$D_G^*(\boldsymbol{x}) = \frac{p_{data}(\boldsymbol{x})}{p_{data}(\boldsymbol{x}) + p_g(\boldsymbol{x})}$$

$$V(G,D) = \int_{\boldsymbol{x}} p_{\text{data}}(\boldsymbol{x}) \log(D(\boldsymbol{x}))dx + \int_{z} p_{\boldsymbol{z}}(\boldsymbol{z}) \log(1 - D(g(\boldsymbol{z})))dz$$
$$= \int_{\boldsymbol{x}} p_{\text{data}}(\boldsymbol{x}) \log(D(\boldsymbol{x})) + p_g(\boldsymbol{x}) \log(1 - D(\boldsymbol{x}))dx$$

Theorem 1. *The global minimum of the virtual training criterion $C(G)$ is achieved if and only if $p_g = p_{data}$. At that point, $C(G)$ achieves the value $-\log 4$.*

$$C(G) = \max_D V(G,D)$$
$$= \mathbb{E}_{\boldsymbol{x}\sim p_{\text{data}}}[\log D_G^*(\boldsymbol{x})] + \mathbb{E}_{\boldsymbol{z}\sim p_{\boldsymbol{z}}}[\log(1 - D_G^*(G(\boldsymbol{z})))]$$
$$= \mathbb{E}_{\boldsymbol{x}\sim p_{\text{data}}}[\log D_G^*(\boldsymbol{x})] + \mathbb{E}_{\boldsymbol{x}\sim p_g}[\log(1 - D_G^*(\boldsymbol{x}))]$$
$$= \mathbb{E}_{\boldsymbol{x}\sim p_{\text{data}}}\left[\log \frac{p_{\text{data}}(\boldsymbol{x})}{P_{\text{data}}(\boldsymbol{x}) + p_g(\boldsymbol{x})}\right] + \mathbb{E}_{\boldsymbol{x}\sim p_g}\left[\log \frac{p_g(\boldsymbol{x})}{p_{\text{data}}(\boldsymbol{x}) + p_g(\boldsymbol{x})}\right]$$

理论分析：收敛性

Proposition 2. *If G and D have enough capacity, and at each step of Algorithm 1, the discriminator is allowed to reach its optimum given G, and p_g is updated so as to improve the criterion*

$$\mathbb{E}_{\boldsymbol{x}\sim p_{data}}[\log D_G^*(\boldsymbol{x})] + \mathbb{E}_{\boldsymbol{x}\sim p_g}[\log(1-D_G^*(\boldsymbol{x}))]$$

then p_g converges to p_{data}

优化的是 θ_g 而非直接优化 p_g，因此 GAN 的收敛性并未得到完全的理论证明

哈爾濱工業大學 HARBIN INSTITUTE OF TECHNOLOGY 成像过程中的底层视觉任务 生成模型 – GAN

实践局限：收敛性

Theorem 1. *The global minimum of the virtual training criterion $C(G)$ is achieved if and only if $p_g = p_{data}$. At that point, $C(G)$ achieves the value $-\log 4$.*

Proof. For $p_g = p_{\text{data}}$, $D_G^*(\boldsymbol{x}) = \frac{1}{2}$, (consider Eq. 2). Hence, by inspecting Eq. 4 at $D_G^*(\boldsymbol{x}) = \frac{1}{2}$, we find $C(G) = \log\frac{1}{2} + \log\frac{1}{2} = -\log 4$. To see that this is the best possible value of $C(G)$, reached only for $p_g = p_{\text{data}}$, observe that

$$\mathbb{E}_{\boldsymbol{x}\sim p_{\text{data}}}[-\log 2] + \mathbb{E}_{\boldsymbol{x}\sim p_g}[-\log 2] = -\log 4$$

and that by subtracting this expression f…

$$C(G) = -\log(4) + \dots \quad (5)$$

where KL is the Kullback–Leibler divergence… …ious expression the Jensen–Shannon divergence between the model's distribution and the data generating process:

$$\boxed{C(G) = -\log(4) + 2\cdot JSD\left(p_{\text{data}}\,\|\,p_g\right)} \quad (6)$$

Since the Jensen–Shannon divergence between two distributions is always non-negative, and zero iff they are equal, we have shown that $C^* = -\log(4)$ is the global minimum of $C(G)$ and that the only solution is $p_g = p_{\text{data}}$, i.e., the generative model perfectly replicating the data distribution. □

Once p_{data} has no overlap with p_g, $JSD(p_{data}\|p_g)$ = **log 2** (constant)

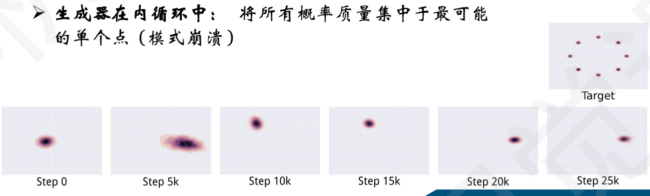
规格严格 功夫到家

哈爾濱工業大學 HARBIN INSTITUTE OF TECHNOLOGY 成像过程中的底层视觉任务 生成模型 – GAN

模式坍塌

$$\min_G \max_D V(G,D) \neq \max_D \min_G V(G,D)$$

- 判别器在内循环中： 收敛到正确的数据分布
- 生成器在内循环中： 将所有概率质量集中于最可能的单个点（模式崩溃）



规格严格 功夫到家

哈爾濱工業大學 HARBIN INSTITUTE OF TECHNOLOGY 成像过程中的底层视觉任务 生成模型 – GAN

Wasserstein GAN (Arjovsky et al.， Arxiv 2017)

- 推图距离

通过李普希兹约束，保证推图距离可计算

$$W(\mathbb{P}_r,\mathbb{P}_\theta) = \sup_{\|f\|_L\le 1} \mathbb{E}_{x\sim\mathbb{P}_r}[f(x)] - \mathbb{E}_{x\sim\mathbb{P}_\theta}[f(x)]$$

for $t = 0, ..., n_{\text{critic}}$ **do**
  Sample $\{x^{(i)}\}_{i=1}^m \sim \mathbb{P}_r$ a batch from the real data.
  Sample $\{z^{(i)}\}_{i=1}^m \sim p(z)$ a batch of prior samples.
  $g_w \leftarrow \nabla_w \left[\frac{1}{m}\sum_{i=1}^m f_w(x^{(i)}) - \frac{1}{m}\sum_{i=1}^m f_w(g_\theta(z^{(i)}))\right]$
  $w \leftarrow w + \alpha\cdot \text{RMSProp}(w, g_w)$
  $w \leftarrow \text{clip}(w, -c, c)$
end for

- L2距离

$$\sup_{\|f\|_L\le 1} \left\|\mathbb{E}_{\boldsymbol{x}\sim p_{\text{data}}}\mathbf{f}(\boldsymbol{x}) - \mathbb{E}_{\boldsymbol{z}\sim p_{\boldsymbol{z}}(\boldsymbol{z})}\mathbf{f}(G(\boldsymbol{z}))\right\|_2^2.$$

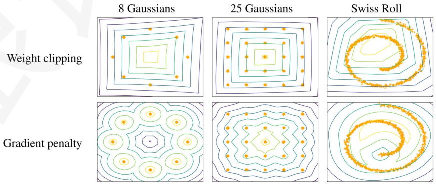
规格严格 功夫到家

哈爾濱工業大學 HARBIN INSTITUTE OF TECHNOLOGY 成像过程中的底层视觉任务 生成模型 – GAN

WGAN-GP

- Gradient penalty

$$L = \underbrace{\mathop{\mathbb{E}}_{\tilde{\boldsymbol{x}}\sim\mathbb{P}_g}[D(\tilde{\boldsymbol{x}})] - \mathop{\mathbb{E}}_{\boldsymbol{x}\sim\mathbb{P}_r}[D(\boldsymbol{x})]}_{\text{Original critic loss}} + \underbrace{\lambda \mathop{\mathbb{E}}_{\hat{\boldsymbol{x}}\sim\mathbb{P}_{\hat{\boldsymbol{x}}}}\left[(\|\nabla_{\hat{\boldsymbol{x}}}D(\hat{\boldsymbol{x}})\|_2 - 1)^2\right]}_{\text{Our gradient penalty}}$$

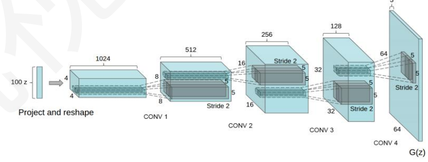


规格严格 功夫到家

哈爾濱工業大學 HARBIN INSTITUTE OF TECHNOLOGY 成像过程中的底层视觉任务 生成模型 – GAN

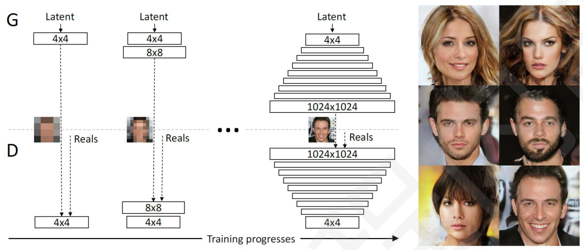
DCGAN (Radford et al., ICLR 2016)

- 全卷积网络
- 在生成器的最后一层和判别器的第一层之外的大部分层中使用 BN
- 判别器的两个batch（真实样本、生成样本）分别进行归一化



规格严格 功夫到家

Progressive GAN (ICLR 2018)



BigGAN (ICLR 2019)

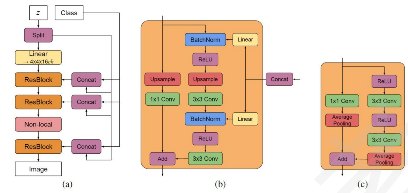
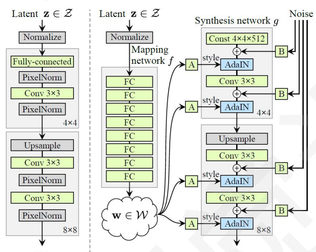


Figure 15: (a) A typical architectural layout for BigGAN's **G**; details are in the following tables. (b) A Residual Block (*ResBlock up*) in BigGAN's **G**. (c) A Residual Block (*ResBlock down*) in BigGAN's **D**.

StyleGAN (Karras, CVPR 2019)

- Progressive GAN
- Multi-layer manipulation on style and noise
- Truncation trick

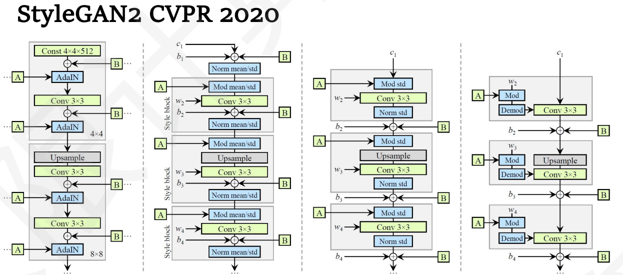


(a) Traditional (b) Style-based generator

StyleGAN2 CVPR 2020

Configuration	FFHQ, 1024×1024 FID ↓	Path length ↓	Precision ↑	Recall ↑	LSUN Car, 512×384 FID ↓	Path length ↓	Precision ↑	Recall ↑
A Baseline StyleGAN [21]	4.40	212.1	**0.721**	0.399	3.27	1484.5	**0.701**	0.435
B + Weight demodulation	4.39	175.4	0.702	0.425	3.04	862.4	0.685	0.488
C + Lazy regularization	4.38	158.0	0.719	0.427	2.83	981.6	0.688	0.493
D + Path length regularization	4.34	**122.5**	0.715	0.418	3.43	651.2	0.697	0.452
E + No growing, new G & D arch.	3.31	124.5	0.705	0.449	3.19	471.2	0.690	0.454
F + Large networks (StyleGAN2)	**2.84**	145.0	0.689	**0.492**	**2.32**	**415.5**	0.678	**0.514**
Config A with large networks	3.98	199.2	0.716	0.422	–	–	–	–

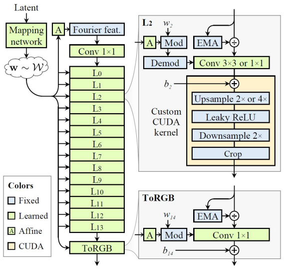
StyleGAN2 CVPR 2020



(a) StyleGAN (b) StyleGAN (detailed) (c) Revised architecture (d) Weight demodulation

StyleGAN3 NIPS2021

StyleGAN3 NIPS2021



	Configuration	FID↓	EQ-T↑	EQ-R↑
A	StyleGAN2	5.14	–	–
B	+ Fourier features	4.79	16.23	10.81
C	+ No noise inputs	4.54	15.81	10.84
D	+ Simplified generator	5.21	19.47	10.41
E	+ Boundaries & upsampling	6.02	24.62	10.97
F	+ Filtered nonlinearities	6.35	30.60	10.81
G	+ Non-critical sampling	4.78	43.90	10.84
H	+ Transformed Fourier features	4.64	45.20	10.61
T	+ Flexible layers (StyleGAN3-T)	4.62	63.01	13.12
R	+ Rotation equiv. (StyleGAN3-R)	**4.50**	**66.65**	**40.48**

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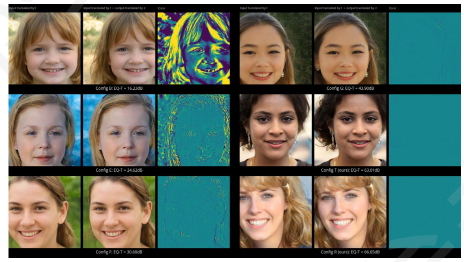

	Configuration	FID↓	EQ-T↑	EQ-R↑
A	StyleGAN2	5.14	–	–
B	+ Fourier features	4.79	16.23	10.81
C	+ No noise inputs	4.54	15.81	10.84
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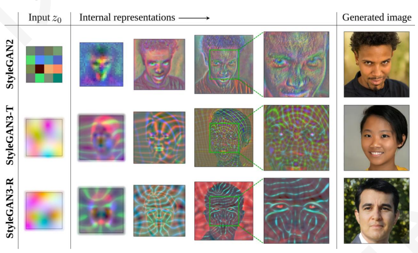

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The End

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